


**KOCAELI UNIVERSITY**

Faculty of Engineering

Department of Electronics and Communication Engineering

# MSP430G2553 BASED SMART ROOM LIGHTING AND ENERGY SAVING SYSTEM

## Project Report

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# ABSTRACT

In this project, a smart room lighting and energy saving system was designed by using the MSP430G2553 microcontroller. The system measures the ambient light level with an LDR sensor and detects human movement with a PIR motion sensor. According to these two inputs, the brightness of an LED output is controlled automatically by a PWM signal . When the environment is dark and motion is detected, the LED is turned on with an appropriate brightness level. When the room is already bright or there is no movement, the LED is turned off. The system status is also displayed on an OLED module and transmitted by using the HC-05 Bluetooth module.

The main purpose of the project is to reduce unnecessary lighting usage while keeping the system simple, low-cost and suitable for real-time embedded control. During the implementation, ADC reading, GPIO interrupt usage, PWM generation, UART communication and I2C based display control were studied together. The final system was tested under different light and motion conditions. The test results showed that the lighting behavior changes correctly according to sensor inputs. The project also provided useful experience about sensor calibration, wiring problems, serial communication and practical debugging on the MSP430 platform.

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# 1. Introduction

This project presents a sensor-based smart lighting system using the MSP430G2553 microcontroller. The system detects movement with a PIR sensor and measures the ambient light level with an LDR sensor. If ther is no movement, the LED remains off. When movement is detected, the LED brightness is adjusted according to the LDR value. The brightness is controlled inversely proportional to the ambient light level; therefore, the LED becomes dimmer in bright environments and brighter in dark environments.

The MSP430G2553 is the main control. The ADC is used to read the LDR value, while the timer module generates the PWM signal for LED brightness control. In addition, an OLED display and HC-05 Bluetooth module are used to show and transmit the system status.

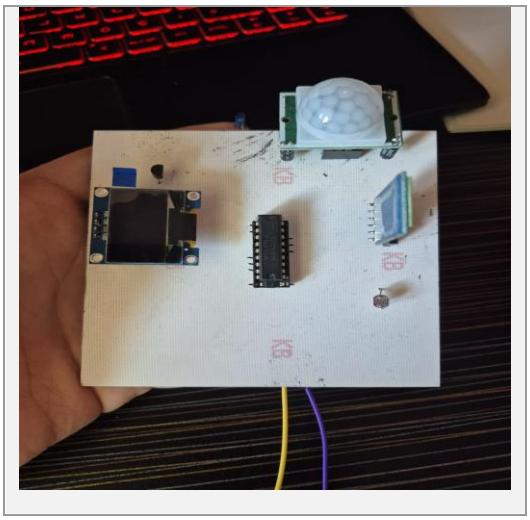


*Figure 1. General view of the completed prototype.*

# 2. Aim and Scope of the Project

The main aim of this project is to design and implement an automatic lighting system that reduces unnecessary LED usage by combining light sensing and motion sensing.

The project scope includes the following tasks:

- Reading the ambient light level with an LDR sensor.
- Detecting human movement with a PIR sensor connected to a digital input pin.
- Generating a PWM control signal by using Timer_A and driving the LED through a 2N2222 NPN transistor.
- Displaying the system status on an OLED display.
- Sending status information through the HC-05 Bluetooth module with UART communication.
- Testing the system in bright, dark, motion and no-motion scenarios.
- Documenting the wiring, software logic, problems and solutions.

## 3. System Overview

The system is built around the MSP430G2553 microcontroller. The LDR sensor gives analog information about the brightness of the environment. This value is converted to a digital value by the ADC. The PIR sensor gives a digital output according to motion. The microcontroller compares the sensor values with predefined conditions and then controls the LED by changing the PWM duty cycle.

The display and Bluetooth module are used for monitoring. The display gives immediate local feedback, while the Bluetooth module allows the same information to be seen on a phone or computer terminal.

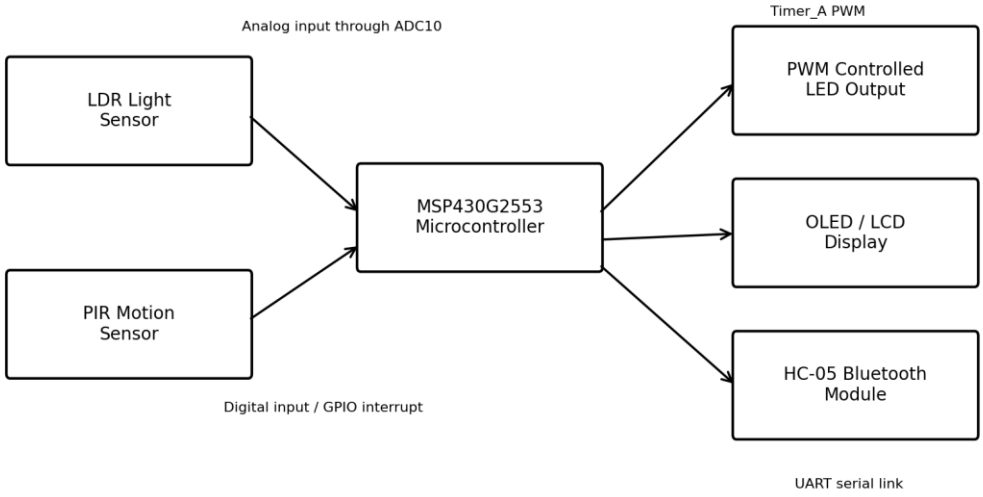



*Figure 2. General block diagram of the system. The LDR and PIR sensors are input units, while the LED, display and Bluetooth module are output/monitoring units controlled by the MSP430G2553.*

The decision logic uses two main questions: Is there motion in the room? Is the room dark enough to require artificial lighting? The LED is activated only when both conditions are satisfied. This prevents the LED from turning on during daytime or when there is no person near the sensor.

## 4. Materials and Components

The hardware of the project consists of a microcontroller board, two sensors, one lighting output, one display module and one wireless communication module. The main components are listed in Table 1.

| **Component** | **Purpose in the project** |
|---|---|
| MSP430G2553 | Main controller; reads sensors and controls outputs |
| LDR sensor | Measures the ambient light level through a voltage divider and ADC input. |
| AMS1117 3.3 V voltage regulator | Provides a stable 3.3 V supply voltage for the circuit. |
| MSP430G2553 LaunchPad | Used for programming and debugging the MSP430G2553. |
| HC-SR501 PIR sensor | Detects human movement |

| | |
|---|---|
| 2N2222 NPN transistor | Works as a switching/driver element for the LED controlled by PWM. |
| SSD1306 1.3” I2C OLED display | Shows motion, light and LED status |
| HC-05 Bluetooth module | Sends serial status messages wirelessly |
| Breadboard and jumper wires | Used for prototype connections |
| 4.7 kΩ, 10 kΩ, 1 kΩ and 100 Ω SMD resistors | Used for pull-up connections, voltage divider, transistor base limiting and LED current limiting. |
| 1 µF, 0.1 µF, 22 µF and 47 µF ceramic capacitors | Used for filtering, decoupling and voltage stabilization in the circuit. |
| 10 × 10 cm single-sided copper clad board | Used for manual PCB fabrication. |

*Table 1. Components used in the project.*

## 4.1 MSP430G2553 Microcontroller

The MSP430G2553 is a low-power 16-bit microcontroller from Texas Instruments. In this project, it is responsible for reading the sensor inputs, generating the PWM control signal, communicating with the OLED display and sending serial data to the Bluetooth module. The PWM signal generated by the microcontroller is applied to a 2N2222 NPN transistor, which is used as the LED driver. Using this microcontroller made the project suitable for studying different embedded system topics in a single application.

The peripherals used in the project are ADC10 for reading the LDR input, GPIO digital input for the PIR sensor, Timer_A for PWM generation, USCI_A0 for UART communication with the HC-05 Bluetooth module and USCI_B0 for I2C communication with the OLED display module.

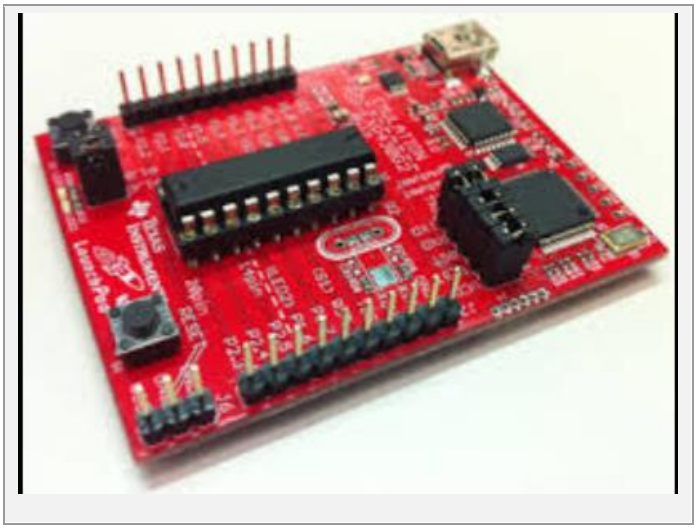


*Figure 3. MSP430G2553 board used as the main control unit.*

## 4.2 LDR Light Sensor

The LDR is a light dependent resistor whose resistance changes according to the amount of light falling on its surface. In this project, the LDR was connected with a 10 kΩ resistor as a voltage divider. The LDR was placed between 3.3 V and the ADC node, while the 10 kΩ resistor was connected between the ADC node and ground. The voltage at the middle point of this divider is connected to the ADC input of the MSP430G2553.

With this connection, the ADC value increases when the environment becomes brighter and decreases when the environment becomes darker. The software uses this ADC value to estimate the ambient light level. If the ADC value is higher than the selected light threshold, the environment is considered bright enough and the LED remains off even if motion is detected. If motion is detected and the ADC value is below the threshold, the LED is activated.

In this case, the PWM duty cycle is adjusted according to the ambient light level so that the LED becomes brighter as the environment becomes darker.

The light threshold and PWM brightness range were selected experimentally. First, the ADC value was observed in a bright environment. Then the LDR was covered or the room was darkened and the ADC value was observed again. These measurements were used to determine a suitable threshold value and brightness control range for the system.

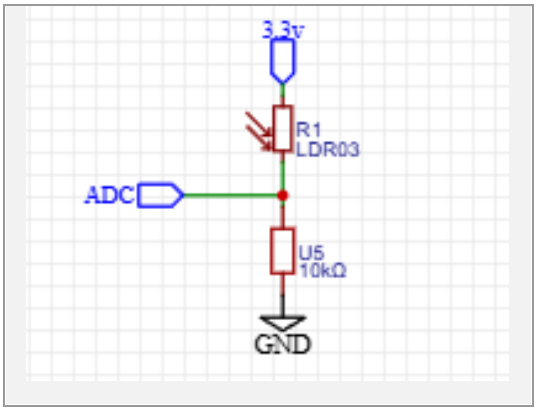



*Figure 4. LDR sensor and voltage divider connection used for ambient light measurement.*

## 4.3 PIR Motion Sensor

The PIR sensor detects changes in infrared radiation caused by human movement. The module output is digital. When motion is detected, the output becomes logic high; otherwise it remains logic low. This makes it easy to connect the module to a digital input pin of the MSP430.

In the project, the PIR output is connected to pin can be read directly in the main loop or configured as an interrupt input. The interrupt approach is useful because the microcontroller can react quickly when motion is detected.

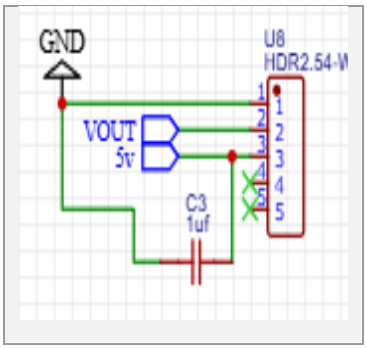



*Figure 5. PIR motion sensor connection for detecting human presence.*

## 4.4 PWM Controlled LED Output

The LED is the lighting output of the system. In this project, the LED is not driven directly from the MSP430G2553 pin. Instead, the PWM signal generated by Timer_A is applied to the base of a 2N2222 NPN transistor through a resistor. The transistor works as a driver and switching element, and it controls the LED current according to the PWM signal.

The LED operation depends on both the PIR sensor and the LDR value. If no motion is detected, the LED remains off. If motion is detected, the software checks the ambient light level using the ADC value from the LDR circuit. When the environment is brighter than the selected threshold, the LED remains off. When the environment is darker than the threshold, the LED is activated and its brightness is controlled by changing the PWM duty cycle. As the environment becomes darker, the duty cycle is increased and the LED becomes brighter.

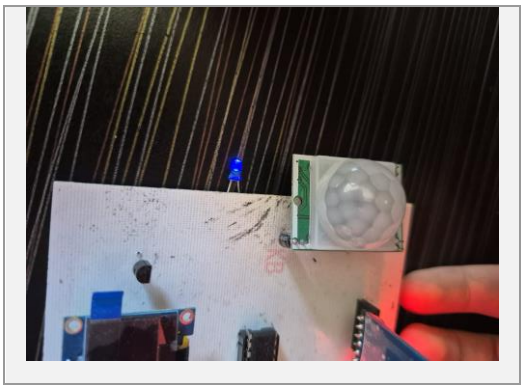


*Figure 6. LED lighting output .*

## 4.5 OLED Display

The display module gives local feedback about the system. During testing, it can show messages such as Motion: YES, Room: DARK and LED: ON. This is helpful because it confirms that the software decision matches the physical output.




*Figure 7. OLED display output used for local system monitoring.*

## 4.6 HC-05 Bluetooth Module

The HC-05 Bluetooth module is used to send system status messages wirelessly. The module communicates with the MSP430 over UART. During the tests, a phone Bluetooth terminal application can be used to receive messages.

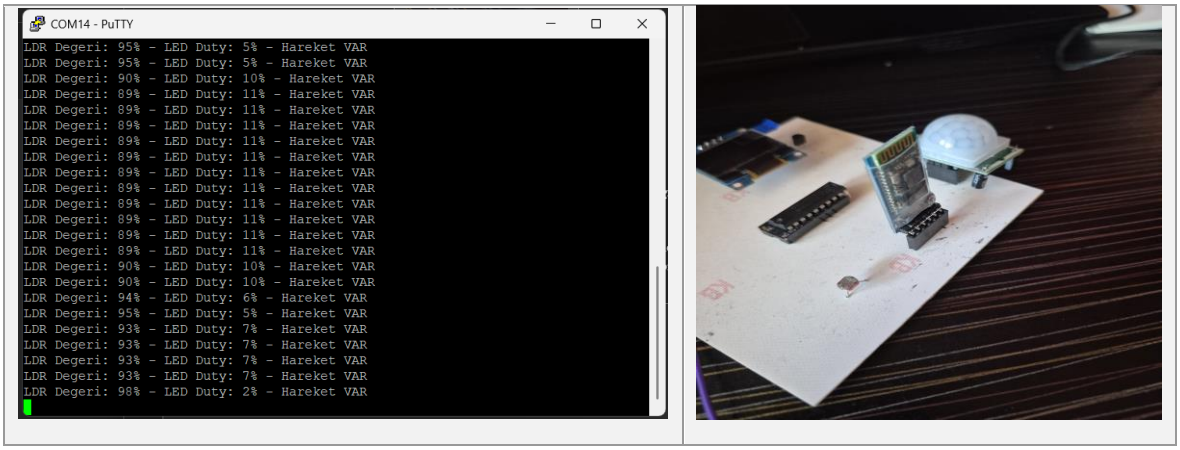



*Figure 8. HC-05 Bluetooth connection and terminal output used for wireless monitoring.*

## 5. Pin Connections and Hardware Interface

| Module / Circuit Part | MSP430 Pin / Net | Peripheral / Function |
|---|---|---|
| LDR sensor voltage divider | P1.0 / A0 | ADC10 analog input |
| PIR motion sensor output | P1.5 / VOUT net | GPIO digital input for motion detection |
| PWM LED driver input | P2.1 / LED_PWM net | Timer_A PWM output |
| 2N2222 transistor base | LED_PWM through 10 kΩ resistor | LED driver / switching control |
| Blue LED | Driven through 2N2222 transistor | Lighting output controlled by PWM |
| OLED display SCL | P1.6 / SCL | I2C clock line |
| OLED display SDA | P1.7 / SDA | I2C data line |
| HC-05 Bluetooth RX/TX | P1.2 / USART_TX and P1.1 / USART_RX | UART communication |
| Reset button | RST pin | Manual reset input |
| AMS1117 voltage regulator | 5 V input / 3.3 V output | Provides regulated 3.3 V supply |
| GND lines | Common GND | Shared reference ground for all modules |
| VCC lines | 3.3 V and 5 V nets | Power supply for MSP430, sensors and modules |

*Table 2. MSP430G2553 pin connections used in the project.*

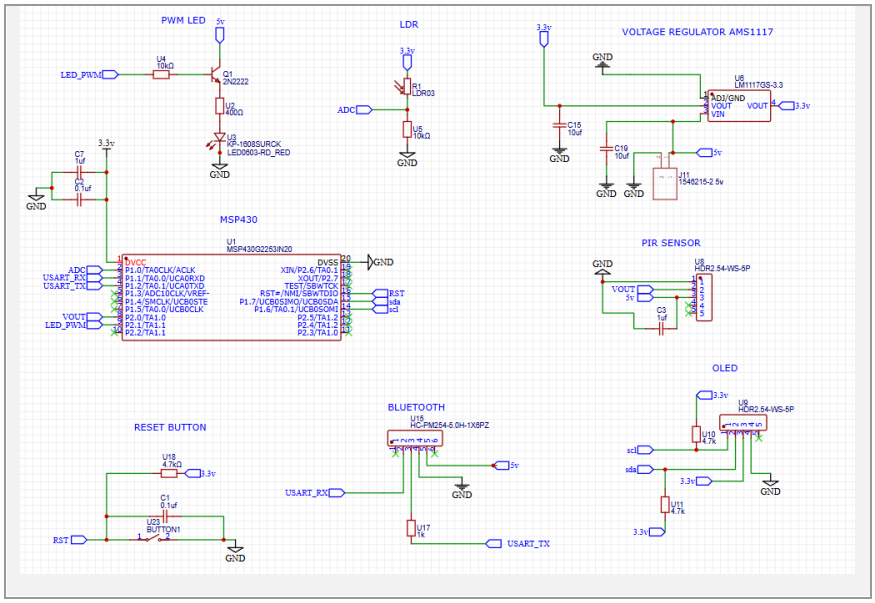



*Figure 9. Complete wiring diagram of the project.*

# 6. Software Design

The software was written with a modular structure. First, the required hardware modules are initialized. After the initialization process, the main loop continuously reads the sensor values and updates the output units. This structure makes the program easier to understand, test and debug.

## 6.1 ADC Reading for LDR

In this project, the LDR is connected between 3.3 V and the ADC node, while a 10 kΩ resistor is connected between the ADC node and ground. With this voltage divider connection, the ADC value increases when the environment becomes brighter and decreases when the environment becomes darker.

The software uses the ADC value to estimate the ambient light level. If the ADC value is higher than the selected light threshold, the environment is considered bright enough and the LED remains off. If the ADC value is lower than the threshold, the environment is considered dark enough for artificial lighting. The threshold value was selected experimentally by observing the ADC values in bright and dark conditions.

Example logic:

```
ldr_value = ADC_Read();


if (ldr_value < LIGHT_THRESHOLD)
    room_state = DARK;
else
    room_state = BRIGHT;
```

## 6.2 PIR Motion Detection

The PIR sensor output is read as a digital value. When the output is logic high, the software accepts that motion is detected. When the output is logic low, the software assumes that there is no motion. This motion information is used together with the LDR reading to decide whether the LED should be activated or not.

The PIR sensor may keep its output high for a short time after movement. This behavior is normal for many PIR modules and should be considered during testing.

Example logic:

```
if (PIR_INPUT == 1)
    motion_state = MOTION_DETECTED;
else
    motion_state = NO_MOTION;
```

## 6.3 PWM Brightness Control with 2N2222 Transistor

Timer_A is used to generate the PWM signal for LED brightness control. The LED is not driven directly from the MSP430G2553 pin. Instead, the PWM signal is applied to the base of a 2N2222 NPN transistor through a resistor. The transistor works as a driver and switching element for the LED.

The LED is activated only when motion is detected and the ambient light level is below the selected threshold. If there is no motion, the LED remains off. If the environment is brighter than the threshold, the LED also remains off even if motion is detected. When motion is detected and the environment is dark enough, the PWM duty

cycle is adjusted according to the LDR value. As the environment becomes darker, the duty cycle is increased and the LED becomes brighter.

Example logic:

```
if (motion_state == MOTION_DETECTED && ldr_value < LIGHT_THRESHOLD)
{
   duty = map_ldr_to_pwm(ldr_value);
   PWM_SetDuty(duty);
}
else
{
   PWM_SetDuty(0);
}
```

### 6.4 Display and Bluetooth Messages

The display and Bluetooth outputs are used to monitor the system state. The OLED display shows short status messages such as motion state, light condition and LED state. The HC-05 Bluetooth module sends similar information as serial text to a phone or computer terminal. This is useful during testing because the sensor readings and software decisions can be observed without using an oscilloscope or logic analyzer.

## 7. Operating Algorithm

7. Operating Algorithm

The operating algorithm is based on a simple and reliable control rule. The LED is allowed to turn on only when motion is detected and the ambient light level is below the selected threshold. This prevents unnecessary activation of the LED in bright environments or when there is no person near the sensor. The same logic can also be written as pseudocode, which is more suitable for a report than inserting the whole source code.

Pseudocode of the system:

Start

Initialize clock, ADC, GPIO, PWM, UART and display

while system is powered:

Read LDR value

Read PIR state

if motion is detected:

if ambient light level is below the threshold:

Adjust LED brightness with PWM according to the LDR value

Show "Motion: YES / LED: ON" on display

Send Bluetooth status message

else:

Turn LED off

Show "Room is bright / LED: OFF" on display

Send Bluetooth status message

else:

Turn LED off

Show "No motion / LED: OFF" on display

Send Bluetooth status message

Repeat loop

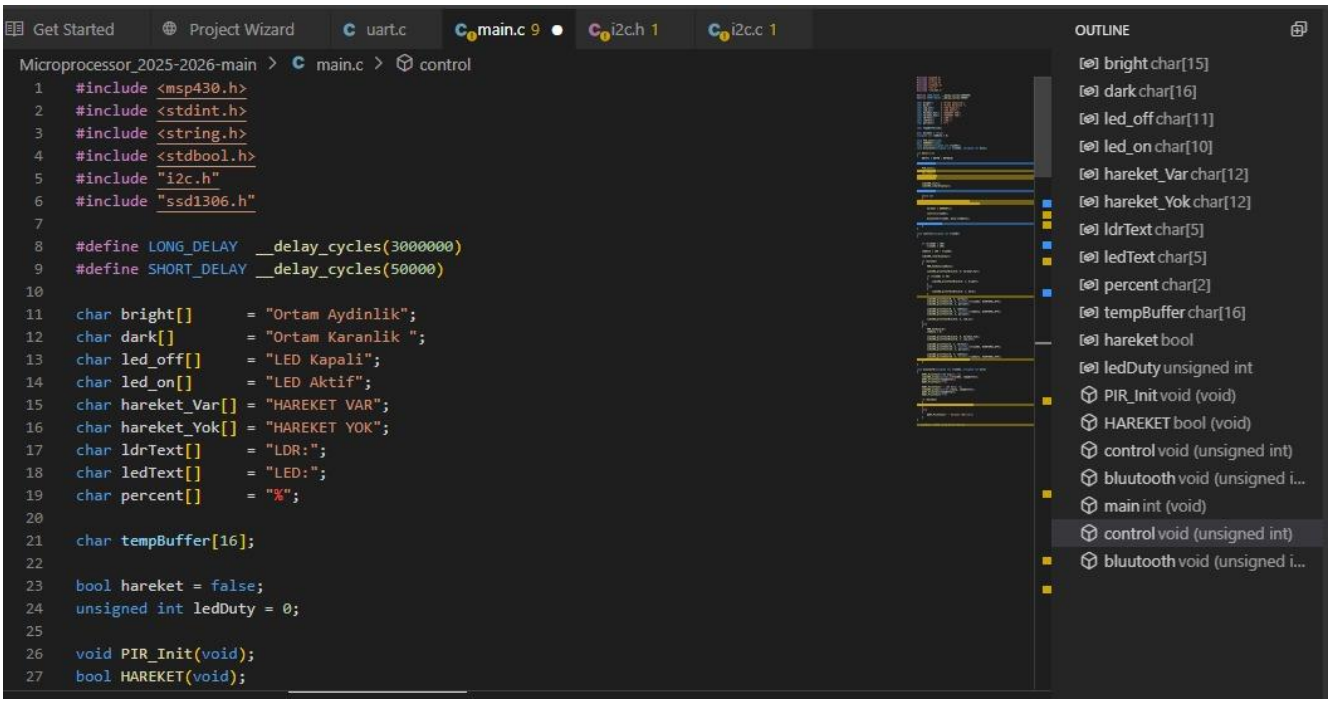



*Figure 10. Development environment screenshot showing the software structure and important code blocks.*

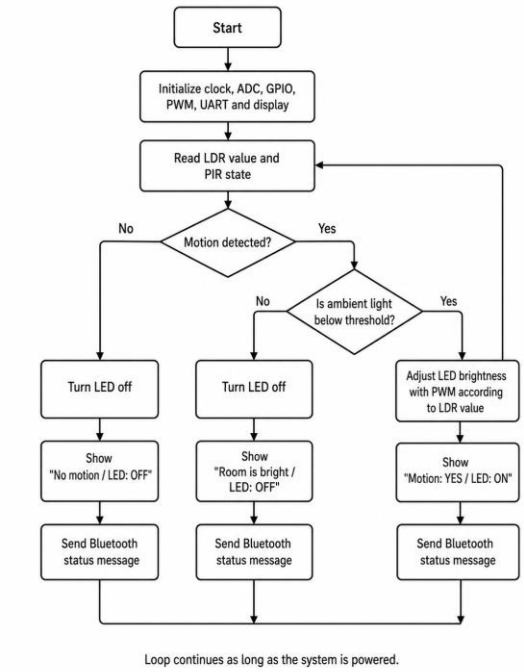



*Figure 11. Flowchart of the smart lighting algorithm. The LED is activated only when both motion and darkness conditions are satisfied.*

## 8. Hardware Implementation

The prototype was first assembled on a breadboard. This allowed fast changes during debugging. The MSP430 board was placed at the center of the circuit. Sensor modules were connected to the input pins, while the LED, display and Bluetooth module were connected to output or communication pins.

After wiring, each part of the circuit was tested separately. First, the LDR value was observed. Then the PIR output was checked. After that, PWM LED control, display output and Bluetooth communication were tested. This step-by-step testing method made it easier to find mistakes because the whole system was not debugged at once.

## 9. Tests and Experimental Results

The system was tested under different room conditions. The purpose of these tests was to confirm that the LED does not turn on unnecessarily and that the output changes according to both sensor inputs. Table 3 summarizes the main test cases.

| **Test case** | **Room light** | **Motion state** | **Expected LED behavior** | **Observed result** |
|---|---|---|---|---|
| 1 | Bright | Motion detected | LED off or low | System prevented unnecessary lighting |
| 2 | Dark | Motion detected | LED on / bright | LED turned on with PWM control |
| 3 | Dark | No motion | LED off | LED remained off |
| 4 | Bright | No motion | LED off | LED remained off |

*Q313Table 3. Test scenarios and observed system behavior.*

## 9.1 Bright Room Test

In the bright room test, the LDR sensed that the environment already had enough light. Even when movement was detected by the PIR sensor, the LED did not need to turn on. This behavior is important because it directly supports the energy saving aim of the project.

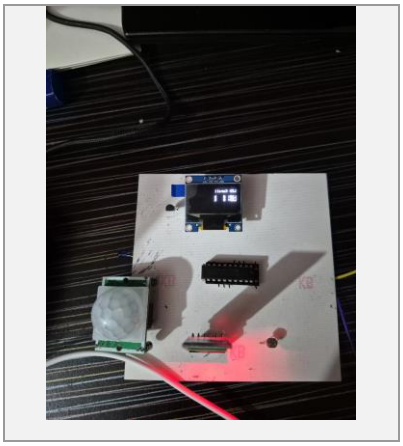


*Figure 12. Bright room test condition. The LED should remain off or at a low level because artificial lighting is not needed.*

## 9.2 Dark Room and Motion Test

In the dark room and motion test, the LDR value passed the darkness threshold and the PIR sensor detected motion. Under these conditions, the PWM output activated the LED. This was the main successful operating condition of the project.



*Figure 13. Dark room with motion test. The LED is activated because both control conditions are satisfied.*

## 9.3 No Motion Test

In the no motion test, the PIR sensor output was low. Even when the room was dark, the LED remained off. This confirms that the system does not depend only on darkness. Human presence is also required for lighting activation.

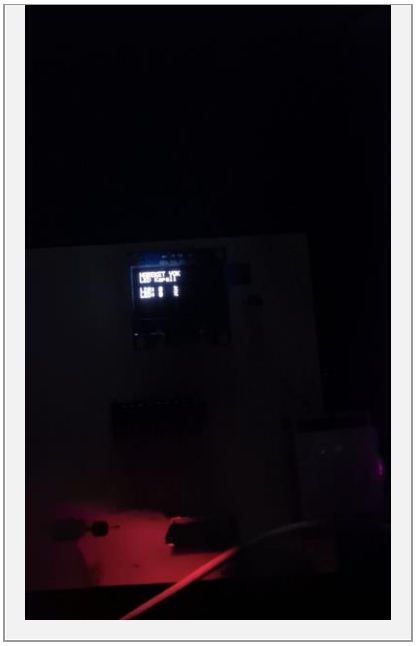


*Figure 14. No motion test result. The lighting output is disabled when human movement is not detected.*

## 9.4 Bluetooth Communication Test

Bluetooth communication was tested with a serial terminal application. The HC-05 module received UART messages from the MSP430 and transmitted them to the phone. This confirmed that the wireless monitoring part of the system was working.

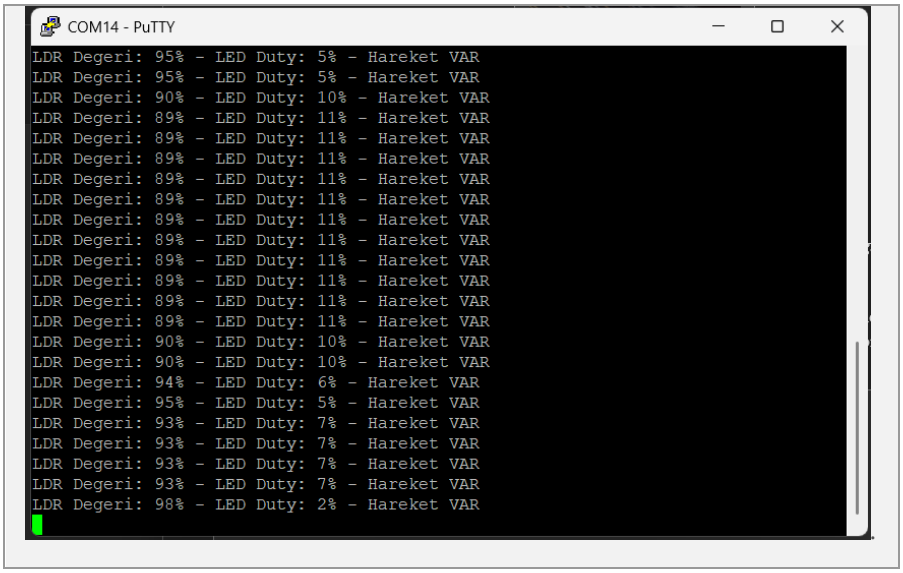



*Figure 17. Bluetooth terminal output showing the status messages sent by the MSP430.*

## 10. Problems Encountered and Solutions

During the project, several practical problems were encountered. These problems were useful because they showed the difference between theoretical circuit design and real hardware implementation.

### 10.1 Display Communication Problem

At the beginning, the display did not show the expected text. The possible reasons were checked one by one: supply voltage, ground connection, I2C address, SDA/SCL wiring and pull-up resistors. After checking the wiring carefully, the display communication became stable. This problem showed that I2C lines must be connected correctly and that the module address should not be assumed without testing.

### 10.2 Bluetooth TX/RX Connection

Another common problem was related to UART wiring. The TX pin of one device must be connected to the RX pin of the other device. If TX is connected to TX or RX to RX, data cannot be received correctly. The baud rate on the MSP430 side and the terminal application also had to be the same.

### 10.3 LDR Threshold Selection

The LDR value changed depending on the room, light direction and sensor position. A fixed threshold did not always behave perfectly in every environment. The solution was to observe ADC values in both bright and dark conditions and select a threshold experimentally. In a future version, the threshold can be made adaptive.

### 10.4 PIR Sensor Delay

The PIR sensor did not always change immediately because many PIR modules include internal timing behavior. This caused short delays while testing. Instead of treating this as a software error, the delay was considered as a normal sensor characteristic. The test procedure was adjusted accordingly.

## 11. Energy Saving Evaluation

The energy saving behavior of the system comes from two conditions. First, the LED does not turn on when there is no movement. Second, the LED does not turn on when the room is bright enough. In a normal manual lighting system, the user may forget to switch the light off. In this project, the software continuously checks the sensor states and disables the output when lighting is not required.

PWM control also makes the design more flexible. Instead of using only full brightness, the LED can be driven with different duty cycle values. This means that the system can provide enough light without always using the maximum output level. Although the prototype uses a small LED, the same control principle can be applied to larger LED drivers with proper power electronics.

## References

[1] Texas Instruments, MSP430G2553 Datasheet.

[2] Texas Instruments, MSP430x2xx Family User Guide.

[3] HC-05 Bluetooth Module Datasheet.

[4] PIR Motion Sensor Module Datasheet.

[5] SSD1306 OLED Display Datasheet.

[6] General application notes for LDR voltage divider circuits and PWM LED control.